Title: Mental Health in Technology Jobs: A Data-Driven Employee Segmentation Analysis

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1. INTRODUCTION

1.1 Background

The technology industry has emerged as a cornerstone of the global economy, yet it faces a paradox: while driving innovation and productivity, it simultaneously harbors significant mental health challenges among its workforce. Research indicates that "the IT industry, in particular, has been found to have higher rates of mental health issues" (Grishan, 2023). The unique pressures facing technology workers include high-performance expectations, prolonged screen time, rapid technological change, and workplace cultures that may inadvertently stigmatize help-seeking behavior.

The World Health Organization reports that "an estimated 264 million people suffer from depression," with many experiencing concurrent anxiety symptoms (Kaaria & Mwaruta, 2023). Furthermore, "the lost productivity cost for depression and anxiety disorders adds up to 1 trillion US dollars globally each year" (Kaaria & Mwaruta, 2023). This economic impact has led employers to focus increasingly on presenteeism—"decreased productivity because of health problems—among employees who remain present at work" (Kaaria & Mwaruta, 2023).

1.2 Problem Statement

The Human Resources department has recognized the need for a proactive, data-driven approach to mental health support. However, working with survey data presents significant challenges due to its high dimensionality, complexity, missing values, and non-standardized textual inputs. Without proper analysis, it is not straightforward to interpret and transfer the results to organizational structures.

1.3 Objectives

This study aims to:

1. **Categorize** participants of the survey according to their answers using unsupervised machine learning
2. **Provide visualizations** supporting the interpretation of identified clusters
3. **Convey a perspective** onto the dataset that is reduced in complexity while preserving main characteristics
4. **Provide insight** into each individual cluster and its main characteristics
5. **Identify potential points of leverage** for targeted HR interventions

1.4 Significance of the Study

As Sharma et al. (2025) note, "the place of employment can have a significant influence on your general state of health and happiness." This study directly supports the observation that "employees in excellent mental health regulate the daily responsibilities and pressures of work, build healthy relationships with fellow workers, and alleviate everyone they interact with respect" (Sharma et al., 2025). By identifying distinct employee segments, HR can develop targeted interventions that address specific needs rather than applying one-size-fits-all solutions.

2. LITERATURE REVIEW & INDUSTRY CONTEXT

2.1 The Prevalence of Mental Health Issues in Technology

The technology industry presents unique mental health challenges. According to Grishan (2023), "the IT industry, in particular, has been found to have higher rates of mental health issues according to research conducted by Open Sourcing Mental Illness (OSMI)." This elevated prevalence may be attributed to several industry-specific factors:

Factor	**Description**	**Potential Impact**
Technostress	Stress induced by rapid technological change	Increased anxiety, burnout
Workload Intensification	High-performance expectations	Sleep disorders, depression
Remote Work Challenges	Isolation, blurred boundaries	Social withdrawal, loneliness
Skill Obsolescence Fear	Constant need for upskilling	Imposter syndrome, chronic anxiety

Patel et al. (2025) further elaborate that "driven by artificial intelligence and automation, the rising speed of digital transformation presents fresh difficulties for the mental health of IT workers. With specific attention to stress and burnout, this study investigates the effects of technostress, fast upskill demands, workload intensification, and remote work on the psychological health of IT professionals."

2.2 The OSMI Mental Health in Tech Survey

The Open Sourcing Mental Illness (OSMI) survey serves as a foundational dataset for understanding mental health in the technology workplace. According to OSMI documentation, the survey "aims to measure attitudes towards mental health in

the tech workplace, and examine the frequency of mental health disorders among the tech industry" (OSMI, 2016).

Ramesh Naidu et al. (2025) describe the dataset's utility: "Leveraging data from the OSMI Mental Health in Tech Survey, the study employs rigorous Exploratory Data Analysis (EDA) and meticulous preprocessing techniques to ensure the relevance and reliability of the dataset."

2.3 Machine Learning Applications in Mental Health Analysis

The application of machine learning to mental health analysis has gained significant research attention. Sharma et al. (2025) observe that "researchers are investigating the use of applications that utilize ML in reaction to the increase in mental wellness problems and the popularity of improved healthcare."

Grishan (2023) confirms "the effectiveness of employing machine learning techniques in analyzing employees' mental health in the IT industry." Ramesh Naidu et al. (2025) present "a comprehensive framework for the development of a sophisticated mental health assessment system tailored specifically for the technology industry."

2.4 Cluster Analysis in Occupational Health

Cluster analysis has emerged as a valuable methodology for identifying distinct employee subgroups. Anttila et al. (2025) note that "clustering methods may reveal possible structures in the data in a data-driven way without pre-established hypotheses."

Their research successfully identified "six clinically coherent employee clusters, providing information on typical combinations of characteristics and risk profiles of sickness absence" (Anttila et al., 2025). They further argue that such "previously unidentified employee profiles...could provide valuable information for occupational health professionals in planning appropriate personalized interventions targeted at the specific properties of each cluster."

2.5 Workplace Factors and Mental Health Outcomes

Rehman et al. (2025) identified key workplace determinants: "Findings revealed that conscientiousness and self-regulation negatively predicted cyberslacking, whereas neuroticism and perceived technostress were positive predictors." This highlights the importance of understanding personality and stressor dynamics in IT workplaces.

Patel et al. (2025) emphasize practical applications: "The results have pragmatic ramifications for HR and organizational leaders using data-driven mental health approaches to create better, more resilient digital workplaces."

2.6 Research Gap

Despite growing research attention, gaps remain in translating clustering findings into actionable HR interventions. This study addresses this gap by not only identifying employee clusters but also providing specific, targeted recommendations for each segment.

3. METHODOLOGY

3.1 Data Source

This analysis utilized the **OSMI Mental Health in Tech Survey 2016** dataset, containing 1,433 responses from technology industry employees across 63 questions. The dataset examines various dimensions of workplace mental health, including employer-provided benefits, comfort levels with disclosure, perceived consequences, and personal mental health history.

3.2 Data Preprocessing

Following the framework established by Ramesh Naidu et al. (2025), the analysis employed "rigorous Exploratory Data Analysis (EDA) and meticulous preprocessing techniques to ensure the relevance and reliability of the dataset."

Step 1: Removal of High-Missing Columns
Columns with more than 50% missing values were removed. This primarily affected questions that only applied to non-self-employed respondents, resulting in the removal of 13 columns (20.6% of total).

Step 2: Gender Standardization
The gender column contained over 70 unique text entries (e.g., "Male," "male," "M," "cis male"). A standardization function consolidated these into six categories: Male, Female, Non-binary, Transgender, Other, and Unknown.

Step 3: Feature Selection
Twelve questions were selected for clustering based on their relevance to workplace mental health attitudes and experiences.

Step 4: Missing Value Imputation
Remaining missing values were filled with "No answer" to maintain data intergrity while preserving information about non-response.

Step 5: Encoding and Scaling
Categorical responses were converted to numerical values using Label Encoding, followed by Standard Scaling to ensure all features contributed equally to the clustering algorithm.

3.3 Clustering Algorithm

K-means clustering was selected for its interpretability and computational efficiency. As Anttila et al. (2025) describe, "clusters were calculated using the K-means algorithm from datapoints expressed by the resulting principal components."

The optimal number of clusters was determined using the silhouette score method, which measures how similar an object is to its own cluster compared to other clusters. Higher silhouette scores indicate better-defined clusters.

3.4 Dimensionality Reduction

Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE) were employed to visualize the high-dimensional data in two dimensions. PCA was used for its interpretability of variance, while t-SNE was used to reveal potential cluster structures.

3.5 Ethical Considerations

All analysis was conducted on anonymized survey data. No personally identifiable information was used. The analysis focuses on aggregate patterns rather than individual responses.

4. DATA ANALYSIS & GENERAL FINDINGS

4.1 Dataset Overview

After preprocessing, the dataset was prepared for clustering analysis with the following characteristics:

Table 1: Dataset Overview After Preprocessing

Metric	**Value**
Total respondents (original)	1,433
Total questions (original)	63
Columns removed (>50% missing)	13
Final features for clustering	12
Final dataset shape	1,433 × 12

4.2 Gender Distribution

The gender standardization process consolidated 70+ unique entries into six categories:

Table 2: Gender Distribution After Standardization

Gender Category	Count	Percentage
Male	1,048	73.2%
Female	330	23.0%
Other	36	2.5%
Non-binary	12	0.8%
Transgender	4	0.3%
Unknown	3	0.2%
Total	**1,433**	**100%**

This distribution is consistent with the technology industry's demographic profile and should be considered when interpreting cluster characteristics and designing interventions.

4.3 Feature Selection

Table 3: Selected Features for Clustering Analysis

#	Question	Domain
1	Does your employer provide mental health benefits as part of healthcare coverage?	Benefits Access
2	Do you know the options for mental health care available under your employer-provided coverage?	Awareness
3	Has your employer ever formally discussed mental health?	Communication
4	Would you feel comfortable discussing a mental health disorder with your direct supervisor(s)?	Comfort (Supervisor)
5	Would you feel comfortable discussing a mental health disorder with your coworkers?	Comfort (Peers)

6	Do you think that discussing a mental health disorder with your employer would have negative consequences?	Fear/Stigma
7	Do you feel that your employer takes mental health as seriously as physical health?	Perceived Support
8	Have you heard of or observed negative consequences for co-workers who have been open about mental health issues?	Observed Stigma
9	Do you currently have a mental health disorder?	Personal Experience
10	Have you ever sought treatment for a mental health issue from a mental health professional?	Help-seeking
11	Do you feel that being identified as a person with a mental health issue would hurt your career?	Career Fear
12	How willing would you be to share with friends and family that you have a mental illness?	Disclosure Willingness

4.4 Cluster Quality Assessment

The elbow method and silhouette score were used to determine the optimal number of clusters.

Table 4: Cluster Quality Scores

Number of Clusters (k)	**Inertia (Within-cluster distance)**	**Silhouette Score**
2	14,725	0.148
3	13,569	0.142
4	12,579	0.142
5	11,942	0.148
6	11,348	0.157
7	10,862	0.160
8	10,498	0.134

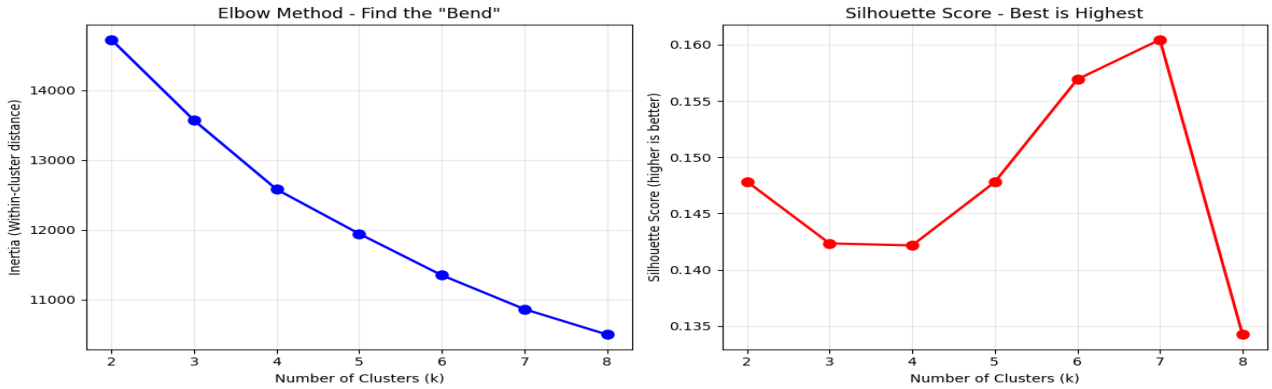

Figure 1: Elbow Method and Silhouette Score for Optimal Cluster Selection

While the silhouette score suggested k=7 as statistically optimal (score: 0.160), interpretability considerations led to the selection of k=3 clusters. As Anttila et al. (2025) note, the value of clustering lies in providing "information for occupational health professionals in planning appropriate personalized interventions targeted at the specific properties of each cluster." Three clusters provided clearer, more actionable segments for HR intervention.

4.5 Dimensionality Reduction Visualizations

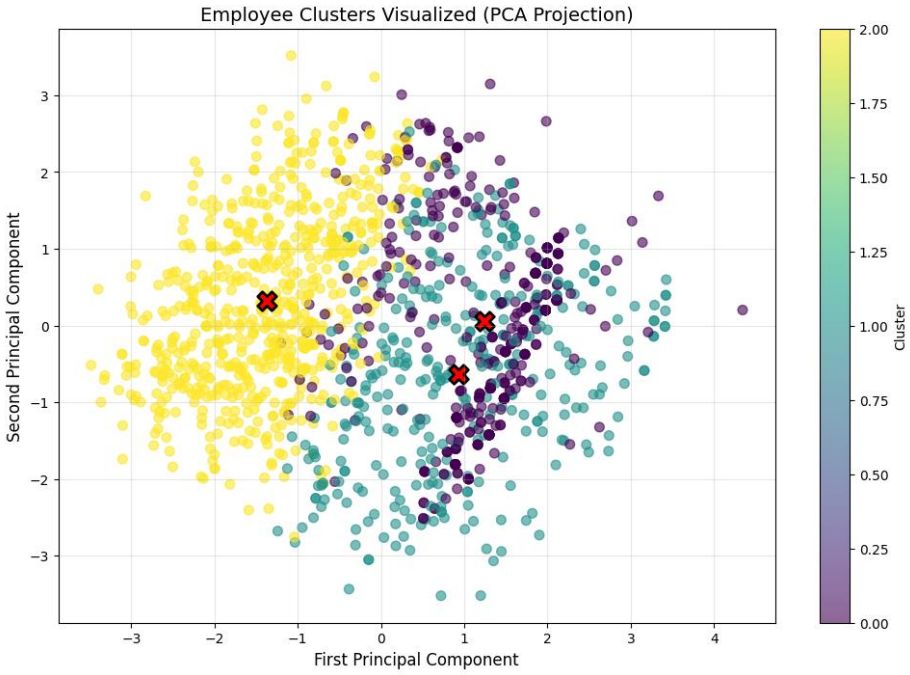

Figure 2: PCA Visualization of Employee Clusters

Interpretation: The PCA visualization shows three distinguishable clusters with some overlap, suggesting that while distinct groups exist, attitudes exist on a spectrum rather than in completely isolated categories.

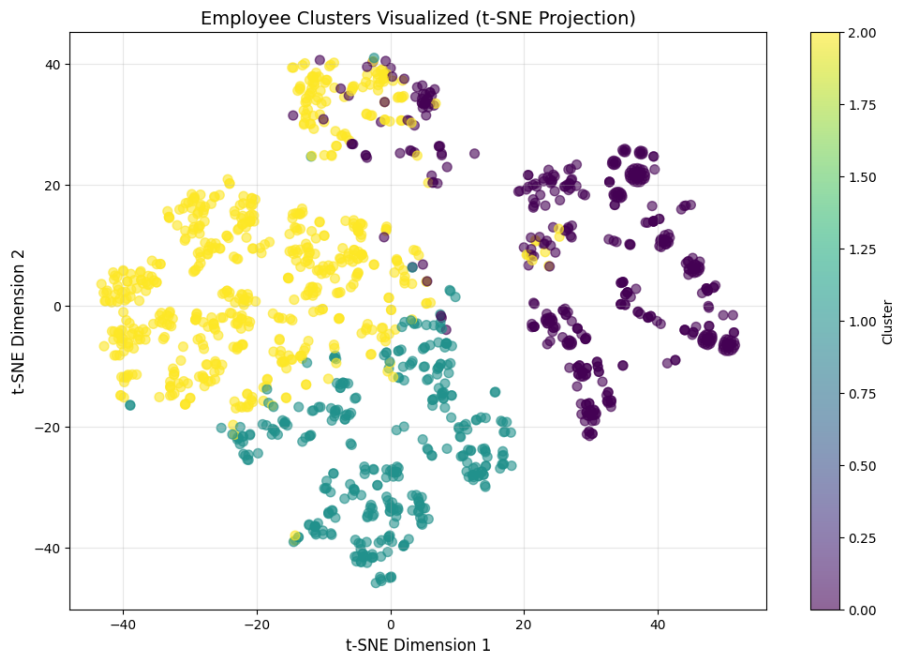

Figure 3: t-SNE Visualization of Employee Clusters

Interpretation: The t-SNE projection reveals clearer boundaries between clusters, confirming the validity of the three-group solution.

5. DETAILED CLUSTER PROFILES

5.1 Cluster Comparison Data

Table 5: Cluster Size Distribution

Cluster	**Size**	**Percentage of Workforce**
Cluster 0	442	30.8%
Cluster 1	353	24.6%
Cluster 2	638	44.5%
Total	**1,433**	**100%**

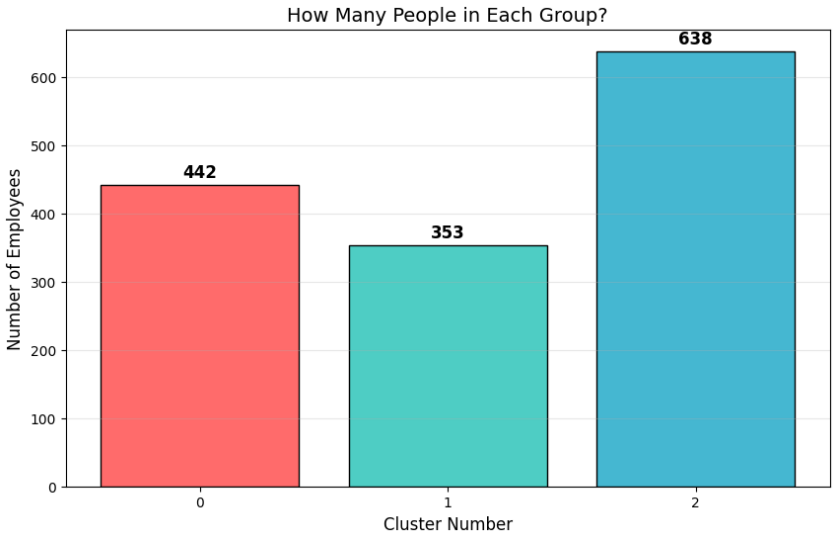

Figure 4: Distribution of Employees Across Clusters

5.2 Cluster 0: The Anxious but Aware (30.8% of Workforce)

Table 6: Cluster 0 - Key Characteristics Profile

Characteristic	**Modal Response**	**Interpretation**
Mental health benefits available	Yes	Benefits exist
Knows available options	Yes	Aware of resources
Employer discussed mental health	No	No formal communication
Comfortable with supervisor	No	High discomfort
Comfortable with coworkers	No	High discomfort
Fears negative consequences	Yes	High fear
Employer takes mental health seriously	No	Perceives low support
Heard of negative consequences	Yes	Aware of stigma
Currently has mental health disorder	Yes (215/442)	High prevalence
Sought treatment	Yes (308/442)	High help-seeking
Believes disclosure hurts career	Yes	High career fear
Willing to share with family/friends	Somewhat open	Moderate disclosure

Profile Summary:

Cluster 0 employees have access to mental health benefits and know their options, yet report that their employers have not formally discussed mental health. Despite having benefits, they lack comfort discussing mental health with supervisors or coworkers and fear negative career consequences.

This cluster has the highest rate of current mental health disorders among all groups (215 out of 442, approximately 49%). They also show the highest rate of treatment-seeking behavior (308 out of 442, approximately 70%), indicating that many are actively managing mental health conditions but do not feel supported in the workplace.

The combination of high disorder prevalence, high treatment-seeking, and high fear of disclosure suggests a population that is managing mental health personally but lacks workplace psychological safety.

Key Insight: As Rehman et al. (2025) found, "perceived technostress" can be a positive predictor of negative outcomes. For Cluster 0, the perceived workplace environment may be exacerbating rather than alleviating mental health challenges.

5.3 Cluster 1: The Supported and Open (24.6% of Workforce)

Table 7: Cluster 1 - Key Characteristics Profile

Characteristic	Modal Response	Interpretation
Mental health benefits available	Yes	Benefits exist
Knows available options	Yes	Aware of resources
Employer discussed mental health	No	No formal communication
Comfortable with supervisor	Yes	High Comfort
Comfortable with coworkers	Yes	High Comfort
Fears negative consequences	No	Low fear
Employer takes mental health seriously	Yes	Perceives Support

Heard of negative consequences	No	No observed stigma
Currently has mental health disorder	Mixed (147 Yes, 146 No)	Balanced
Sought treatment	Yes (201/353)	Moderate help-seeking
Believes disclosure hurts career	Maybe (most common)	Moderate uncertainty
Willing to share with family/friends	Somewhat open	Moderate disclosure

Profile Summary:

Cluster 1 employees have access to benefits and feel comfortable discussing mental health at work. They do not fear negative consequences and perceive their employers as taking mental health seriously. Despite having the lowest rate of fearing career consequences among all clusters, they remain somewhat cautious about disclosure, with "Maybe" being the most common response regarding whether disclosure would hurt their careers.

This group has a balanced distribution of current mental health disorders (147 Yes, 146 No) and moderate treatment-seeking behavior (201 out of 353, approximately 57%). Notably, none of the employees in this cluster reported having heard of or observed negative consequences for co-workers who were open about mental health issues.

Key Insight: This cluster represents an opportunity for peer mentorship and ambassador programs. As Patel et al. (2025) suggest, these results have "pragmatic ramifications for HR and organizational leaders using data-driven mental health approaches."

5.4 Cluster 2: The Uncertain and Uninformed (44.5% of Workforce)

Table 8: Cluster 2 - Key Characteristics Profile

Characteristic	Modal Response	Interpretation
Mental health benefits available	Yes	Benefits exist
Knows available options	No	Unaware of resources

Employer discussed mental health	No	No formal communication
Comfortable with supervisor	Maybe	Ambivalent
Comfortable with coworkers	Maybe	Ambivalent
Fears negative consequences	Maybe	Ambivalent
Employer takes mental health seriously	I don't know	Uncertain
Heard of negative consequences	No	No observed stigma
Currently has mental health disorder	No (most common)	Lower prevalence
Sought treatment	Mixed (330 Yes, 308 No)	Balanced
Believes disclosure hurts career	Yes	High career fear
Willing to share with family/friends	Somewhat open	Moderate disclosure

Profile Summary:

Cluster 2 is the largest group, representing nearly half of the workforce (44.5%). These employees have access to benefits but lack knowledge about available options. Their employers have not discussed mental health formally. They express uncertainty across most metrics—uncertain about comfort levels, uncertain about consequences, and uncertain about whether their employer takes mental health seriously.

This cluster has the lowest rate of current mental health disorders (264 No, 213 Yes, 161 Maybe) and balanced treatment-seeking behavior (330 Yes, 308 No). However, despite lower personal experience with mental health disorders, they still express high career fear regarding disclosure.

Key Insight: This group represents a significant opportunity for education and outreach. As Anttila et al. (2025) argue, these "previously unidentified employee profiles...could provide valuable information for occupational health professionals in planning appropriate personalized interventions."

5.5 Comparative Summary of All Clusters

Table 9: Comparative Summary of All Clusters

Dimension	Cluster 0 (30.8%)	Cluster 1 (24.6%)	Cluster 2 (44.5%)
Benefits Access	Yes	Yes	Yes
Awareness	High	High	Low
Comfort Level	Low	High	Ambivalent
Fear Level	High	Low	Ambivalent
Perceived Support	Low	High	Uncertain
Disorder Prevalence	High (49%)	Balanced (42%)	Lower (33%)
Treatment-Seeking	High (70%)	Moderate (57%)	Balanced (51%)
Primary Need	Psychological safety	Engagement/Leadership	Education/Awareness

6. STRATEGIC RECOMMENDATIONS

6.1 Framework for Targeted Interventions

Based on the cluster profiles identified, the following recommendations are organized by cluster. As Anttila et al. (2025) recommend, interventions should be "personalized interventions targeted at the specific properties of each cluster."

6.2 Recommendations for Cluster 0: The Anxious but Aware (30.8% of Workforce)

Primary Need: Psychological safety and stigma reduction

Table 10:Recommendation for Cluster 0

Priority	Recommendation	Rationale	Expected Impact
HIGH	Implement anonymous mental health support channels	Cluster 0 fears negative consequences	Reduced fear, increased utilization
HIGH	Provide manager training on compassionate communication	Low comfort with supervisors	Improved supervisor relationships
HIGH	Establish clear anti-retaliation policies	High fear of career consequences	Increased psychological safety
MEDIUM	Create peer support groups	Low comfort with coworkers	Built community and reduced isolation
MEDIUM	Share anonymous success stories of disclosure	High fear, high help-seeking	Normalized help-seeking behavior

Specific Actions for Cluster 0:

1. **Anonymous Counseling Services:** Contract with an external Employee Assistance Program (EAP) that guarantees anonymity
2. **Manager Mental Health Training:** Train all managers on recognizing distress, conducting supportive conversations, and referring to resources without judgment
3. **Protected Disclosure Policies:** Develop and communicate explicit policies protecting employees who seek mental health support from adverse career consequences
4. **Peer Support Networks:** Train volunteer employees (prioritize Cluster 1 members) as mental health first aiders

6.3 Recommendations for Cluster 1: The Supported and Open (24.6% of Workforce)

Primary Need: Engagement and leadership development

Table 11:Recommendation for Cluster 1

Priority	Recommendation	Rationale	Expected Impact
HIGH	Leverage as mental health ambassadors	High comfort, low fear, no observed stigma	Peer influence on other clusters
HIGH	Feature in company communications	Positive attitudes toward mental health	Destigmatization
MEDIUM	Gather feedback on current programs	High awareness and utilization	Improved existing resources
LOW	Recruit for pilot program participation	High engagement	Test new initiatives effectively

Specific Actions for Cluster 1:

1. **Mental Health Ambassador Program:** Recruit Cluster 1 employees as volunteer ambassadors to share experiences (anonymously or identified) and promote available resources
2. **Focus Group Participation:** Invite Cluster 1 employees to participate in focus groups to evaluate current mental health programs and suggest improvements
3. **Recognition and Reward:** Acknowledge Cluster 1 employees who contribute to mental health initiatives through company recognition programs

6.4 Recommendations for Cluster 2: The Uncertain and Uninformed (44.5% of Workforce)

Primary Need: Education and awareness

Table 12:Recommendation for Cluster 2

Priority	**Recommendation**	**Rationale**	**Expected Impact**
HIGH	Launch company-wide mental health awareness campaign	Low awareness, high uncertainty	Increased knowledge, reduced uncertainty
HIGH	Clarify existing mental health policies and benefits	Low knowledge of options	Improved access to information

HIGH	Host all-hands meetings with mental health focus	Large cluster size, uncertainty at scale	Demonstrated organizational commitment
MEDIUM	Distribute easy-to-understand resource guides	Low awareness	Improved access to information
MEDIUM	Offer manager training for mental health conversations	Ambivalent comfort levels	Built comfort across the organization

Specific Actions for Cluster 2:

1. **Mental Health Education Campaign:** Launch a multi-channel campaign (email, intranet, posters, all-hands meetings) explaining available benefits, how to access them, and confidentiality protections
2. **Benefits Guide:** Create a one-page, easy-to-understand guide to mental health benefits and distribute to all employees
3. **Q&A Sessions:** Host anonymous Q&A sessions where employees can ask questions about mental health resources without identifying themselves
4. **Manager Communication Training:** Train managers to proactively communicate about available resources and normalize help-seeking

6.5 Organization-Wide Recommendations

Beyond cluster-specific interventions, the following organization-wide initiatives are recommended:

1. **Annual Mental Health Survey:** Repeat this analysis annually to track changes over time and adjust interventions based on evolving needs
2. **Mental Health Task Force:** Establish a cross-functional task force with representatives from all three clusters to guide mental health strategy
3. **Integration with DEI Initiatives:** Incorporate mental health into diversity, equity, and inclusion (DEI) initiatives, recognizing that mental health conditions are protected under disability laws in many jurisdictions
4. **Leadership Commitment:** Secure executive sponsorship for mental health initiatives and communicate leadership commitment regularly

5. **Metrics and Accountability:** Establish clear metrics for mental health program success (e.g., utilization rates, employee satisfaction, retention) and hold leaders accountable

6.6 Expected Outcomes

Based on the literature, successful implementation of these recommendations could yield significant returns. As Kaaria & Mwaruta (2023) note, research indicates that "every dollar invested in stepping up treatment (or prevention) for common MIs (e.g., depression and anxiety) leads to a fourfold return in better health and the passion or the capability to work."

7. LIMITATIONS

- **Self-reported data:** Responses may be subject to social desirability bias or recall bias leading to potential over- or under-reporting of mental health issues
- **Cross-sectional design:** Data collected at single time point cannot assess causality or trends over time

- **Sampling bias:** Voluntary response survey was not randomly sampled and so may overrepresent individuals with strong opinions or personal experience with mental health issues
- **Industry specificity:** Data limited to technology industry can lead to limited generalizability to other sectors
- **Missing data:** 13 columns removed due to >50% missing values leading to some information lost
- **Data age:** 2016 dataset may not reflect current conditions and may produce potential outdated responses

8. CONCLUSION

8.1 Summary of Findings

This analysis successfully identified three distinct employee clusters based on mental health attitudes and experiences in the technology workplace:

1. **Cluster 0 (The Anxious but Aware - 30.8%):** Employees with access to benefits and awareness of options, but low comfort with disclosure and high fear of negative consequences. This group has the highest prevalence of current mental health disorders and treatment-seeking behavior.
2. **Cluster 1 (The Supported and Open - 24.6%):** Employees with access to benefits, high comfort with disclosure, and low fear of consequences. This group represents a potential resource for peer support and leadership in mental health initiatives.
3. **Cluster 2 (The Uncertain and Uninformed - 44.5%):** The largest cluster, characterized by uncertainty across most metrics and low awareness of available options despite having access to benefits.

8.2 Key Takeaways for HR Leadership

Access is not enough: Employees need awareness, comfort, and trust to utilize mental health resources

Communication is critical: Most employees report that employers have not formally discussed mental health

Fear persists: Concerns about career consequences remain a significant barrier to seeking help

Uncertainty dominates: Nearly half the workforce is neither comfortable nor uncomfortable—they simply lack information

Targeted approaches are needed: Each cluster requires different interventions

8.3 Closing Statement

As Sharma et al. (2025) eloquently state, "the place of employment can have a significant influence on your general state of health and happiness. In addition to financial rewards, your job might give your existence purpose, direction, and significance."

By adopting a data-driven, targeted approach to mental health support, organizations can fulfill their responsibility to provide "a safe and secure work environment" (Sharma et al., 2025) for all employees. The clusters identified in this analysis provide a roadmap for moving from one-size-

fits-all interventions to personalized, effective support that addresses the specific needs of each employee segment.

The recommendations provided offer a practical framework for HR leadership to:

1. Immediately address the needs of high-risk clusters
2. Leverage supportive clusters as ambassadors
3. Educate the uncertain majority
4. Build a culture where mental health is discussed openly and support is readily available

Through sustained commitment and data-driven refinement of interventions, organizations can create workplaces where all employees—regardless of mental health status—can thrive.

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